MAT5210 Sample Final Examination

(This is part of the take-home examination of MAT5210 during the pandemic. Difficulty of the exam may vary every year.)

1. Let $\zeta := e^{2\pi i/21}$ and $K = \mathbf{Q}(\zeta)$ be the cyclotomic extension.

   (a) Which abelian group is $\mathrm{Gal}(K/\mathbf{Q})$ isomorphic to? Express your answer in terms of
   $$\mathrm{Gal}(K/\mathbf{Q}) \cong \frac{\mathbf{Z}}{p_1\mathbf{Z}} \times \ldots \times \frac{\mathbf{Z}}{p_k\mathbf{Z}},$$
   where $p_1 \leq \cdots \leq p_k$ are prime numbers.

   (b) List all subgroups of $\mathrm{Gal}(K/\mathbf{Q})$ and their inclusion relations.

   (c) It is known that
   - $\sqrt{-3} = \zeta^7 - \zeta^{-7}$,
   - $\sqrt{-7} = -(\zeta^3 - \zeta^{-3})(\zeta^6 - \zeta^{-6})(\zeta^9 - \zeta^{-9})$,
   - $\cos\left(\frac{2\pi}{7}\right) = \zeta^3 + \zeta^{-3}$

   Write down all intermediate fields between K and $\mathbf{Q}$ using the above expressions, and match them with the subgroups obtained in (b) under the Galois correspondence.

2. Determine the ideal class groups (not just their orders) of $\mathbf{Z}[\sqrt{-14}]$ and $\mathbf{Z}[\sqrt{-21}]$.

3. Using the fact that the class number of $\mathbf{Z}[\sqrt{-5}]$ is 2, show that the equation
   $$x^3 = y^2 + 5$$
   has no integer solutions.

4. (Harder Question) Let $L \supset K \supset \mathbf{Q}$ be number fields, and $\mathcal{O}_L \supset \mathcal{O}_K \supset \mathbf{Z}$ be their rings of integers. In this question, we will define $N_{L/K}$, the norm of an ideal of L over K.

   (a) Let P_L be any prime ideal of $\mathcal{O}_L$, prove that $P_K := P_L \cap \mathcal{O}_K$ is a prime ideal in $\mathcal{O}_K$.

(b) Show that $P_L \mid P_K \cdot \mathcal{O}_L$, i.e., P_L lies above P_K.

Then the norm of any nonzero prime ideal P_L over K is given by $N_{L/K}(P_L) := P_K^f$, where $f := [\mathcal{O}_L/P_L : \mathcal{O}_K/P_K]$ is the index of $\mathcal{O}_L/P_L$ as a $\mathcal{O}_K/P_K$-module.

(c) In the special case when $K = \mathbf{Q}$ and P_L lies above the prime ideal $p\mathbf{Z}$ in $\mathbf{Z}$ with $N(P_L) := |\mathcal{O}_L/P_L| = p^f$ for some integer f as defined in the lectures. Show that $N_{L/\mathbf{Q}}(P_L) = (p\mathbf{Z})^f$. In other words, our new definition of $N_{L/\mathbf{Q}}$ generalizes the norm N defined in the lectures.

Assume $L : \mathbf{Q}$ is Galois from now on (and hence $L : K$ is also Galois). Then for any prime ideal P_L of $\mathcal{O}_L$, we saw in the lectures that $P_K \cdot \mathcal{O}_L = (S_1 \cdots S_l)^e$ for some prime ideals S_i of $\mathcal{O}_L$ with $S_1 = P_L$.

(d) Show that $N_{L/K}(P_L) \cdot \mathcal{O}_L = (S_1 \cdots S_l)^{ef} = \prod_{\sigma \in \mathrm{Gal}(L/K)} \sigma(P_L)$.

More generally, by unique factorization of prime ideals, the norm of any nonzero ideal I_L of $\mathcal{O}_L$ is defined by $\mathcal{I}_L = Q_1 Q_2 \cdots Q_s \Rightarrow N_{L/K}(I_L) := N_{L/K}(Q_1) N_{L/K}(Q_2) \cdots N_{L/K}(Q_s)$, where Q_i are all prime ideals of $\mathcal{O}_L$.

(e) Using (d), show that $N_{L/K}(I_L) = \prod_{\sigma \in \mathrm{Gal}(L/K)} \sigma(I_L)$ for any ideal I_L of $\mathcal{O}_L$.

(f) Let $\zeta \in \mathcal{O}_L$. Recall the definition of the norm of an element is given by $\mathrm{Norm}_{L/K}(\zeta) := \Pi_{\sigma \in \mathrm{Gal}(L/K)} \sigma(\zeta) \in \mathcal{O}_K$. Show that $\mathrm{Norm}_{L/K}(\zeta) \cdot \mathcal{O}_L = N_{L/K}(\zeta \cdot \mathcal{O}_L) \cdot \mathcal{O}_L$ and hence $\mathrm{Norm}_{L/K}(\zeta) \cdot \mathcal{O}_K = N_{L/K}(\zeta \cdot \mathcal{O}_L)$ (recall that when $K = \mathbf{Q}$, we have proved in class that $\mathrm{Norm}_{L/\mathbf{Q}}(a) = N\big((a)\big)$).